

Darrick Aaron Untarman

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SUMMARY

IoT-focused IT student with 5 months of experience as an IoT Engineer Intern, building end-to-end systems integrating embedded hardware, real-time data processing, and cloud platforms. Experienced in building full-stack IoT solutions using AWS Cloud, ROS 2, and embedded hardware to solve real-world problems through hackathons and projects.

EDUCATION

Asia Pacific University

2023 - 2026

Bachelor of Science (Hons) in Information Technology (Internet of Things) | CGPA: 3.67

EXPERIENCE

HSIL Hackathon 7th Edition – UCSI Hospital (CMET)

April 2026

Project: MediGuard (Non-Contact Patient Monitoring System)

- Built MediGuard in a team of 6, a non-contact patient monitoring system using mmWave radar and face detection with AWS cloud integration (S3, DynamoDB, Lambda, API Gateway) and web dashboard.
- Enabled real-time monitoring of multiple vital signs (heart rate, breathing rate), improving hygiene and supporting safer patient observation in healthcare environments.

IoT Engineer Intern – Iotech Solutions

June 2025 - November 2025

- Contributed to 3+ IoT development projects during internship, including flood monitoring, AMR automation, and wearable data solutions involving hardware, firmware, and cloud integration.
- Performed circuit design, firmware development, sensor calibration, and system testing for reliable IoT deployments.

AWS Hackathon Finalist – AMR Medical Inventory System

October 2025

- Developed an autonomous mobile robot system in a team-based hackathon environment, integrating ROS 2 and SLAM navigation with AWS cloud for real-time inventory tracking.
- Implemented robotic medicine delivery and cloud-based monitoring dashboard for improved hospital logistics efficiency.
- Designed an AI-assisted medicine dispensing safety system using a camera with YOLO-based object detection to verify medication correctness before enabling drawer access, reducing risk of incorrect medicine dispensing.

PROJECTS

Smart Flood Monitoring Weather Station

- Developed a real-time monitoring system using ESP32, multiple sensors, LoRa communication, and Firebase cloud with mobile dashboard visualization using Flutter.
- Enabled remote tracking of environmental data (temperature, rainfall, water level) for early flood detection and faster response awareness.

ADDITIONAL INFORMATION

Technical skills:

- Programming:** Python (Intermediate), Java (Basic), PHP (Basic), SQL (Basic), Flutter (Basic)
- Web Development:** HTML, CSS, JavaScript (Intermediate)
- IoT & Cloud:** AWS, Firebase, Node-RED, Raspberry Pi, MQTT, ESP32, Arduino (Intermediate)
- AI, Robotics & Engineering:** YOLOv8, ROS 2, SLAM, SolidWorks (Basic)

Language: English (Professional), Mandarin (Conversational), Bahasa Melayu (Conversational), Hokkien (Native)

Certification: AWS Academy Graduate - Cloud Architecting - Training Badge, Vice Chancellor's (VC) List for Academic Excellence (2024), Cisco (Network Security, Introduction to Networking)